Teacher answer key: Primary 1, Semester 1 (DRAFT)

DRAFT: this answer key is for owner review. Do not give it to pupils. Contact: pypteltd@gmail.com (business email for now, to be changed later)

This key lists the answers and the result of the check on every page. You do not need it for routine marking: a pupil who has a working check has the right answers. To spot-check, compare the check strip with the line "Check result" below.

How the topic tags work

Each entry names the syllabus outcomes the puzzle practises, with the outcome numbers of the Primary 1 content in the Mathematics Syllabus, Primary (2021 edition, pages 31 and 32). The references are for your planning only and are not printed on the pupil pages. "Problem solving" marks a page whose questions come from the kinds found in school tests rather than from one numbered outcome.

Pages in this booklet (Terms 1 and 2)

Strand	Pages
Number and Algebra - Whole Numbers	S1-01, S1-02, S1-03, S1-06, S1-08, S1-10, S1-12, S1-13, S1-14, S1-15
Measurement and Geometry - Geometry	S1-04, S1-05, S1-16, S1-17
Number and Algebra - Addition and Subtraction	S1-07, S1-09, S1-11
Measurement and Geometry - Measurement	S1-18, S1-19
Number and Algebra - Multiplication and Division	S1-20

The page code in the footer, for example S1-07 (P1-08), gives the page in this booklet and, in brackets, its number in the full set of 40.

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S1-01 (P1-25) Count the Crowd 1 star

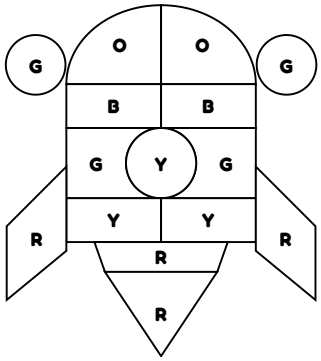
Topic: Counting a set and comparing two sets

Syllabus outcomes: Whole Numbers 1.1, Whole Numbers 1.4. Notes on the tag: 1.1, 1.4 (counting to tell the number of objects in a given set; comparing the number of objects in two sets)
Puzzle type: Code-breaker (counting). How the pupil checks: Code word (5 letters).
Answers: 1) 7, 2) 10, 3) 10 + 5 = 15, 4) 6 - 5 = 1 more, 5) 5 - 2 = 3 fewer
Check result: Code word: CLOTH
Teacher note: Three counting items (the third is a full ten frame and some more) and two comparing items. In the comparing items every picture in the shorter row is joined by a line to one in the longer row; the pictures with no line are the answer to how many more or how many fewer. Pictures are objects, never bare dots. Every set is small; the ten frame item is a full ten and some more.

S1-02 (P1-01) Colour the Rocket 1 star

Topic: Place value: tens and ones

Syllabus outcomes: Whole Numbers 1.2. Notes on the tag: 1.2 (tens and ones)
Puzzle type: Colour by code (pupils colour with their own crayons). How the pupil checks: Colour key and tally: every answer must be on the key, the number of parts of each colour must match the box, and the parts in the same place on the left and on the right match.
Answers: Colour key: 1 = yellow, 2 = red, 3 = blue, 4 = green, 5 = orange
Check result: Tally: yellow 3, red 4, blue 2, green 4, orange 2; the parts in the same place on the left and on the right match.



Teacher note: Every part asks a tens-and-ones question: how many ones or tens a number has, or the missing part of a number made of tens and ones. The answers are 1 to 5. Pupils colour the black-and-white outline with their own crayons (or, for a pencil-only copy, write Y, R, B, G or O in each part: the tally and the matching parts can be checked the same way). Check strip: count the parts of each colour (the tally is printed); the parts in the same place on the left and on the right match.

S1-03 (P1-24) Words and Numbers 1 star

Topic: Number words, numerals and ordinals

Syllabus outcomes: Whole Numbers 1.3, Whole Numbers 1.7. Notes on the tag: 1.3, 1.7 (reading and writing numbers in numerals and in words; ordinal numbers and symbols)
Puzzle type: Maze / path. How the pupil checks: Route: each step goes to a box whose top word says the number at the foot of the box you leave; the line must reach FINISH through the right number of boxes.
Answers: Route (row, column from top-left): (1,1) ninety-five / 17 -> (1,2) seventeen / 9th -> (2,2) ninth / 31 -> (2,3) thirty-one / 4th -> (1,3) fourth / 29 -> (1,4) twenty-nine / 94 -> (2,4) ninety-four / 38 -> (3,4) thirty-eight / 5th
Check result: Route reaches FINISH through 8 boxes; every top word names the number at the foot of the box before it.

1	2	5	6
	3	4	7
			8

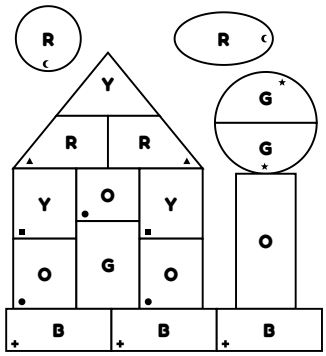
Teacher note: A 4 by 4 grid. Each box has a number word on top and a numeral at the foot. From a box, the pupil moves to a touching box whose top word says the numeral at the foot of the box they are leaving. 2 of the 7 steps are ordinals (a word such as ninth against its symbol 9th); the rest are numbers to 99 in words and numerals. Wrong turns are boxes that look right but are not: digits swapped (35 and fifty-three), teen and ty mixed up (13 and thirty), an ordinal against its plain number. 2 wrong turns can be entered from the route and lead nowhere, so the box count and FINISH catch them.

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S1-04 (P1-12) Colour the House 1 star

Topic: Shapes: corners

Syllabus outcomes: 2D Shapes 1.1. Notes on the tag: Geometry 1.1 (identify, name, describe 2D shapes: triangle, square, rectangle, circle)
Puzzle type: Colour by code (pupils colour with their own crayons). How the pupil checks: Colour key and tally: every answer must be on the key, the number of parts of each colour must match the box, and parts with the same small mark match.
Answers: Colour key: 3 = yellow, 4 = red, 6 = blue, 7 = green, 8 = orange
Check result: Tally: yellow 3, red 4, blue 3, green 3, orange 4; parts with the same small mark match.



Teacher note: Shapes are triangle, square, rectangle and circle only. A circle has no corners. Pupils colour the black-and-white outline with their own crayons (or, for a pencil-only copy, write Y, R, B, G or O in each part: the tally and the matching parts can be checked the same way). Check strip: count the parts of each colour (the tally is printed); parts with the same small mark match.

S1-05 (P1-19) Shape Count 1 star

Topic: 2D shapes: rectangle, triangle, circle, half circle, quarter circle

Syllabus outcomes: 2D Shapes 1.1, 2D Shapes 1.3. Notes on the tag: Geometry 1.1, 1.3 (name 2D shapes; identify the shapes in a figure)
Puzzle type: Count + code-breaker. How the pupil checks: The five counts add to 15, and a code word (5 letters).
Answers: rectangles: 2, triangles: 4, circles: 5, half circles: 3, quarter circles: 1 (total 15)
Check result: Total 15; code word: BUILD
Teacher note: The picture has no squares so there is no 'is a square a rectangle?' question. Shapes are turned in different directions on purpose. The five counts are all different (1 to 5) so each count gives one letter.

S1-06 (P1-07) The Queue 2 stars

Topic: Ordinal numbers: 1st to 5th, next to

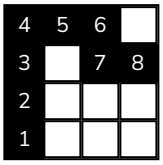
Syllabus outcomes: Whole Numbers 1.7. Notes on the tag: 1.7 (ordinal numbers first to tenth)
Puzzle type: Logic (queue). How the pupil checks: Shirt letters read from 1st to 5th spell a word (5 letters; one child is not in the queue).
Answers: 1st Jenny, 2nd Aisha, 3rd Ali, 4th Ravi, 5th Lily; Kumar is not in the queue (letter W is the decoy)
Check result: Word: ENJOY
Teacher note: Six children are shown but the queue has five places: a clue says who is not in the queue, and that child's shirt letter (W) is a decoy that is not part of the word. The top row is not in queue order. The five letters of the word have exactly one arrangement that is a word, so no wrong queue spells a word: every wrong queue of five of the six children was tested against a very large word list. Clues on this page: ordinal numbers (3rd, 5th), next to. Two or more clues compare two children, so the pupil must combine them. Syllabus ordinals go to 10th; this page uses 1st to 5th.

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S1-07 (P1-08) Make Twelve 1 star

Topic: Addition within 20

Syllabus outcomes: Addition and Subtraction 2.1, Addition and Subtraction 2.2, Addition and Subtraction 2.7.
Notes on the tag: 2.1 to 2.7 (add within 20, mental calculation)
Puzzle type: Maze / path. How the pupil checks: Route: the line must reach FINISH through the right number of boxes, every box making 12.
Answers: Route (row, column from top-left): (4,1) 8+4 -> (3,1) 5+7 -> (2,1) 9+3 -> (1,1) 6+6 -> (1,2) 6+6 -> (1,3) 5+7 -> (2,3) 9+3 -> (2,4) 4+8
Check result: Route reaches FINISH through 8 boxes; every box on it adds to 12.



Teacher note: A 4 by 4 grid: sixteen sums, all within 20. Boxes off the route make 8, 10, 11, 14 and a slip in adding sends the pupil there. 2 extra boxes that do make 12 are dead ends. Counting the boxes shows a line that takes a shortcut.

S1-08 (P1-37) Bonds and Families 2 stars

Topic: Number bonds to 20: parts, wholes and the loop that closes

Syllabus outcomes: Addition and Subtraction 2.1, Addition and Subtraction 2.2, Addition and Subtraction 2.3.
Notes on the tag: 2.1, 2.2, 2.3 (concepts of addition and subtraction; use of +, - and =; relationship between addition and subtraction)
Puzzle type: Chain loop (bond tiles). How the pupil checks: Loop: the number in an empty circle must match the circle it is joined to by =, all the way round and back to tile 1.
Answers: Empty circles: tile 1 left = 3, tile 2 left = 5, tile 3 left = 10, tile 4 right = 3, tile 5 left = 3, tile 6 left = 7. Tiles (whole = left + right): 8 = 3 + 5, 15 = 5 + 10, 14 = 10 + 4, 7 = 4 + 3, 10 = 3 + 7, 10 = 7 + 3
Check result: Every touching pair matches, all the way round.
Teacher note: Six tiles, each a whole (the top circle) with two parts. One part of every tile is empty; the pupil finds it from the whole and the other part (adding up to the whole, or taking the part away from it). The right part of a tile is the same number as the left part of the next tile, and the right part of the last tile is the first tile's left part. So every number the pupil writes is tested against a matching circle: a wrong number breaks the loop. The syllabus asks for the relationship between addition and subtraction; the pupil uses it each time a part is found from the whole. The plan also had a list of pairs that make 10 and a second sentence for each tile: both were dropped because a self-check at 95% could not be made for them.

S1-09 (P1-10) Take Away Code 1 star

Topic: Subtraction within 20; missing numbers

Syllabus outcomes: Addition and Subtraction 2.1, Addition and Subtraction 2.3, Addition and Subtraction 2.7.
Notes on the tag: 2.1 to 2.7 (subtract within 20, inverse relationship)
Puzzle type: Code-breaker. How the pupil checks: Code word (6 letters) and putting the number back.
Answers: 1) 9, 2) 7, 3) 4, 4) 8, 5) 5, 6) 16
Check result: Code word: COUSIN
Teacher note: Items 5 and 6 ask for the number taken away. Pupils can check by adding back, which is the inverse idea.

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S1-10 (P1-21) Story Sums 2 stars

Topic: Add and take away stories within 20

Syllabus outcomes: Addition and Subtraction 2.1, Addition and Subtraction 2.2, Addition and Subtraction 2.7. Notes on the tag: 2.1, 2.2, 2.7 (concepts of addition and subtraction, +, - and =, mental calculation within 20)
Puzzle type: Code-breaker (stories). How the pupil checks: Code word (6 letters).
Answers: 1) 3 + 3 = 6, 2) 3 + 2 = 5, 3) 6 + 6 = 12, 4) 6 - 4 = 2, 5) 6 - 3 = 3, 6) 5 + 9 = 14
Check result: Code word: MONKEY
Teacher note: Six short stories within 20, each with a picture of the first group. The pupil chooses add or take away from the story. One story says 'fewer than' about the other child: the child asked about has more, so the answer is an addition. Where possible the table decoys are the answers a pupil gets by using the wrong operation, and any other value has no letter at all, so a wrong operation that is written down never spells the word. A box left blank is not seen by the check: every box must get a letter. The stories use more than and fewer than and 'how many more'.

S1-11 (P1-09) Magic Square 2 stars

Topic: Adding three 1-digit numbers

Syllabus outcomes: Addition and Subtraction 2.4. Notes on the tag: 2.4 (adding more than two 1-digit numbers)
Puzzle type: Magic square. How the pupil checks: All eight line totals make 15.
Answers: Rows: 4 9 2 / 3 5 7 / 8 1 6
Check result: All 8 totals = 15

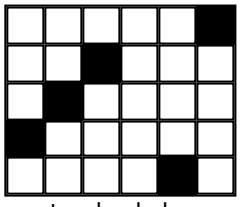
4	9	2
3	5	7
8	1	6

Teacher note: Only three numbers are given. The solver shows the other six are forced. The sum boxes let a pupil see which line is wrong. Each number 1 to 9 must be used once.

S1-12 (P1-38) More or Fewer 2 stars

Topic: Comparing amounts: more than, fewer than and making the same

Syllabus outcomes: Whole Numbers 1.5, Addition and Subtraction 2.1 (problem solving). Notes on the tag: 1.5, 2.1 (comparing and ordering numbers; concepts of addition and subtraction; make equal and fewer than: problem solving)
Puzzle type: Logic grid. How the pupil checks: Letters beside the amounts, read down the list, spell a word (5 letters; one of the six amounts is not used).
Answers: Ali: 20 (G), Mei: 8 (L), Siti: 7 (O), Ben: 5 (R), Kumar: 19 (Y); nobody has 16 (letter M is the decoy)
Check result: Letters down the list: GLORY



Teacher note: Six amounts are listed but there are only five children: a clue says which amount nobody has, and its letter is a decoy that is not part of the word. The clues are exact (a number more than, a number fewer than) and one clue says what a child gives so that two children have the same. The five letters of the word have exactly one arrangement that is a word, so no wrong placement spells a word: every wrong way to give five children five of the six amounts was tested against a very large word list. A fewer than clue is the other way round from a more than clue, so the pupil must read who has the greater amount. The page shows a worked example of crosses and ticks.

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S1-13 (P1-02) Mystery Numbers 2 stars

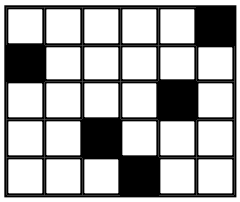
Topic: Numbers to 100: tens digit, ones digit, more and less

Syllabus outcomes: Whole Numbers 1.2, Whole Numbers 1.5. Notes on the tag: 1.2, 1.5 (tens and ones, compare and order numbers)
Puzzle type: Number riddles. How the pupil checks: Code word (5 letters) and testing each clue.
Answers: 1) 66, 2) 85, 3) 36, 4) 56, 5) 26
Check result: Code word: JUICE
Teacher note: Every riddle has 3 clues and every clue is needed. A pupil who stops after one clue gets more than one possible number; testing the other clues removes them. Clues used on this page: more than, less than, ones digit, tens digit.

S1-14 (P1-04) Sticker Grid 2 stars

Topic: Comparing numbers to 100 (more, fewer, greatest, smallest)

Syllabus outcomes: Whole Numbers 1.5. Notes on the tag: 1.5 (compare and order numbers)
Puzzle type: Logic grid. How the pupil checks: Letters beside the numbers, read down the list, spell a word (5 letters; one of the six numbers is not used).
Answers: Ali: 99 (F), Mei: 12 (R), Siti: 88 (U), Ben: 63 (I), Kumar: 78 (T); nobody has 23 (letter P is the decoy)
Check result: Letters down the list: FRUIT



Teacher note: Six numbers are listed but there are only five children: a clue says which number nobody has, and its letter (P) is a decoy that is not part of the word. The five letters of the word have exactly one arrangement that is a word, so no wrong placement of the numbers spells a word: every wrong way to give the children five of the six numbers was tested against a very large word list. A pupil whose letters are not a word must go back to the clues. The page shows a worked example of crosses and ticks. Clues on this page: more than, fewer than, greatest, second greatest.

S1-15 (P1-26) Chart Patterns 2 stars

Topic: Number patterns on a chart: 10 more, 10 less, and a growing table

Syllabus outcomes: Whole Numbers 1.6, Whole Numbers 1.2, Whole Numbers 1.5. Notes on the tag: 1.6, 1.2, 1.5 (patterns in number sequences; tens and ones; comparing and ordering numbers)
Puzzle type: Code-breaker (chart and pattern). How the pupil checks: Code word (6 letters).
Answers: 1) 43, 2) 76, 3) 70, 4) 44, 5) 9, 6) 15
Check result: Code word: PEANUT
Teacher note: A piece of a 100 chart with two boxes, two questions about 10 more and 10 less (one box down or up the chart), and a growing table: the sweets go up by 3 each day. The two table answers continue the pattern. The syllabus asks for patterns in number sequences; the chart makes the 10 more, 10 less pattern visible. The plan drew this page as a hidden picture; a hidden picture can reject only about two thirds of wrong numbers, so it is a code word page to meet the 95% rule.

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S1-16 (P1-33) Shape Builder 2 stars

Topic: 2D shapes: count the shapes in a figure and form a figure

Syllabus outcomes: 2D Shapes 1.1, 2D Shapes 1.2, 2D Shapes 1.3. Notes on the tag: Geometry 1.1, 1.2, 1.3 (naming and classifying 2D shapes; forming a figure from rectangle, square, triangle, half circle, quarter circle; the shapes in a figure)
Puzzle type: Count + form a figure + code-breaker. How the pupil checks: The five counts add to 18; their letters and the letter of the pieces that form the small figure spell a word (6 letters).
Answers: rectangle: 4, square: 6, triangle: 2, half circle: 1, quarter circle: 5 (total 18); the small figure is made of rectangle and half circle (letter E)
Check result: Total 18; code word: BUDGIE
Teacher note: The big figure is built from the five shapes of the syllabus list: squares and rectangles (straight sides only) and triangles, half circles and quarter circles (the half and quarter circles have a curved side). Pieces touch edge to edge and some are turned. The five counts are all different, so each count has one letter. The table says which shapes have straight sides and which have a curved side. The small figure task is forming a figure: the pupil sees an outline and picks which two pieces fit together to make it; the three sets have different total areas, so only one set can form the outline. Its letter is the sixth letter of the word.

S1-17 (P1-34) Finish the Figure 2 stars

Topic: Copy figures on a square grid

Syllabus outcomes: 2D Shapes 1.4. Notes on the tag: Geometry 1.4 (copying figures on a dot grid or square grid)
Puzzle type: Copy + matching pictures. How the pupil checks: The pupil finishes each copy, then picks the printed right half that matches; the four letters spell a word (4 letters).
Answers: 1) option A (letter F), 2) option B (letter R), 3) option A (letter O), 4) option A (letter G)
Check result: Letters in order: FROG
Teacher note: Each figure is drawn on a square grid with straight lines along the grid and across some squares. The left half of each copy is already drawn; the pupil copies the right half from the figure above it. Then the pupil compares their drawing with three printed right halves (each lettered) and writes the letter of the one that is the same. The wrong halves differ by one square or one slanting line. No wrong pick of one letter in each row spells a word. The page cannot see the pupil's drawing: it checks that the pupil can tell which printed half matches the figure.

S1-18 (P1-31) Shortest to Longest 1 star

Topic: Length in cm: measure and put in order

Syllabus outcomes: Length 1.2, Length 1.3. Notes on the tag: Length 1.2, 1.3 (use of abbreviation cm; comparing and ordering lengths in cm)
Puzzle type: Logic (strips in order). How the pupil checks: Letters of the five shortest strips, read from the shortest to the longest, spell a word (5 letters; the longest strip is a decoy).
Answers: Lengths: T = 11 cm, B = 2 cm, N = 9 cm, L = 3 cm, K = 10 cm, A = 6 cm. Shortest to longest: B L A N K (longest strip T is the decoy)
Check result: Letters, shortest first: BLANK
Teacher note: Six strips, each starting at an arrow (some at 0, some not). The pupil counts the cm to the other end, puts the strips in order from shortest to longest and writes the letters, leaving out the longest. The five letters of the word have exactly one arrangement that is a word, so no wrong order spells a word: every wrong way to write five of the six letters was tested against a very large word list. The cm length boxes are for the pupil to work in; the check is the letters. The syllabus asks for lengths in cm; unit strips (paper clips) used in school practice are not used on this page.

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S1-19 (P1-17) How Long? 1 star

Topic: Length: measure in centimetres

Syllabus outcomes: Length 1.1, Length 1.2. Notes on the tag: Length 1.1, 1.2 (measure in cm, abbreviation cm)
Puzzle type: Code-breaker. How the pupil checks: Code word (6 letters).
Answers: 1) 4 cm, 2) 5 cm, 3) 3 cm, 4) 2 cm, 5) 7 cm, 6) 8 cm
Check result: Code word: LENGTH
Teacher note: A ruler is drawn under each strip at 10 mm to 1 cm. If the page is copied at a different scale the strip and ruler change together, so the reading from the drawn ruler stays right, but a real ruler laid on the page will not match: print at 100% (actual size) if pupils will use a real ruler. Two strips start at 0 and four start at 1 to 4 on the ruler; a bold arrow and a bold tick show the start. The Primary 1 length outcomes list measuring in cm and do not list measuring from a mark other than 0, so the offset strips are a small extension: the pupil counts the cm from the arrow to the end of the strip. Every length is a whole number of cm.

S1-20 (P1-13) Equal Groups 1 star

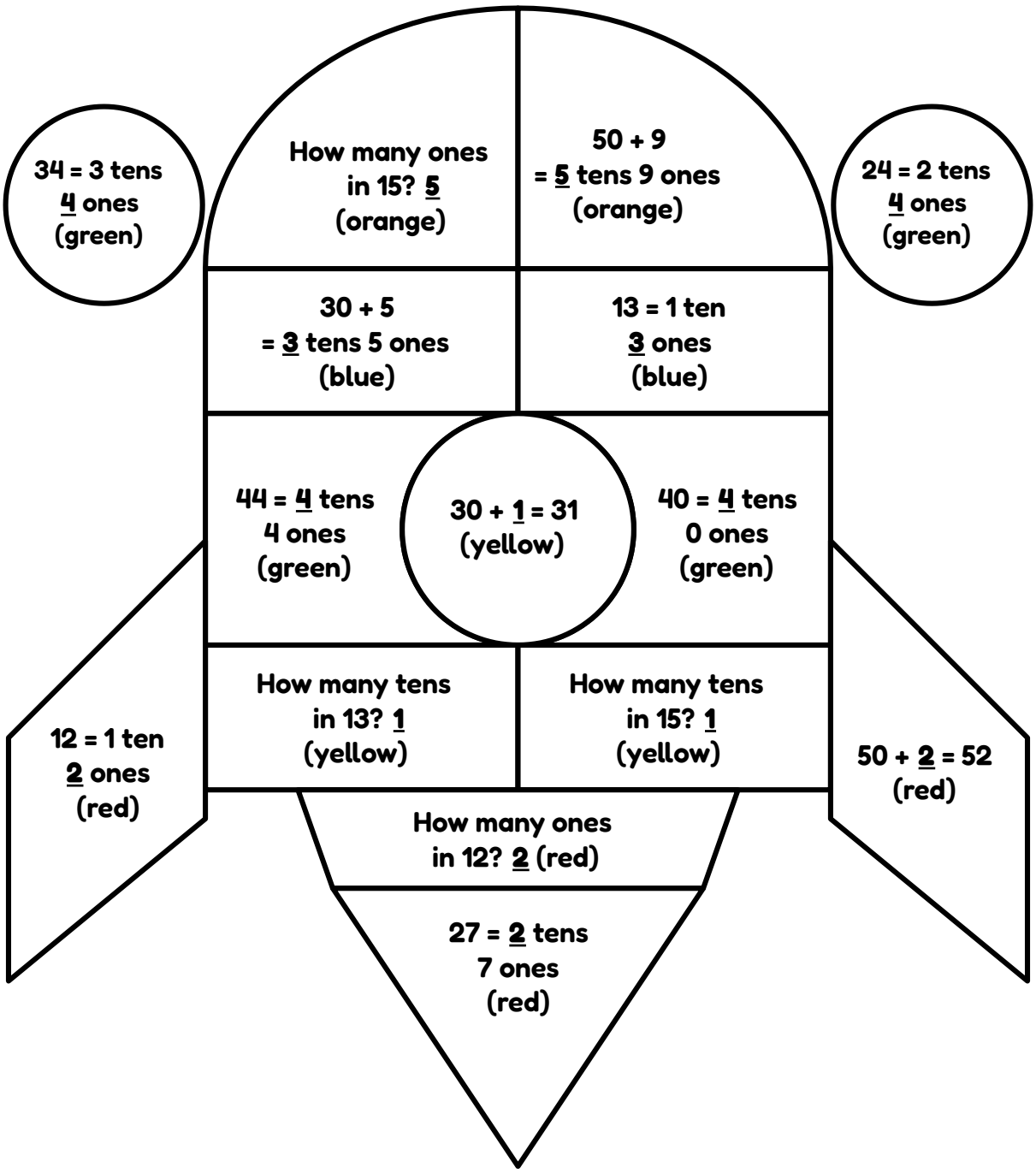
Topic: Multiplication: equal groups and repeated addition

Syllabus outcomes: Multiplication and Division 3.1, Multiplication and Division 3.2, Multiplication and Division 3.3. Notes on the tag: 3.1 to 3.3 (concept of multiplication, sign x, multiplying within 40)
Puzzle type: Code-breaker. How the pupil checks: Code word (6 letters).
Answers: 1) 4 bags of 2 = 8, 2) 3 bags of 5 = 15, 3) 3 bags of 2 = 6, 4) 4 bags of 4 = 16, 5) 3 bags of 4 = 12, 6) 3 bags of 3 = 9
Check result: Code word: NUMBER
Teacher note: Pictures of equal groups. Pupils write the number of groups, the number in each group and the total. Totals are within 16 and the objects are drawn large (about 8 mm). Repeated addition (4 bags of 3 is 3 + 3 + 3 + 3) is the bridge to the sharing pages in the other semester.

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Answers: S1-02 (P1-01) Colour the Rocket

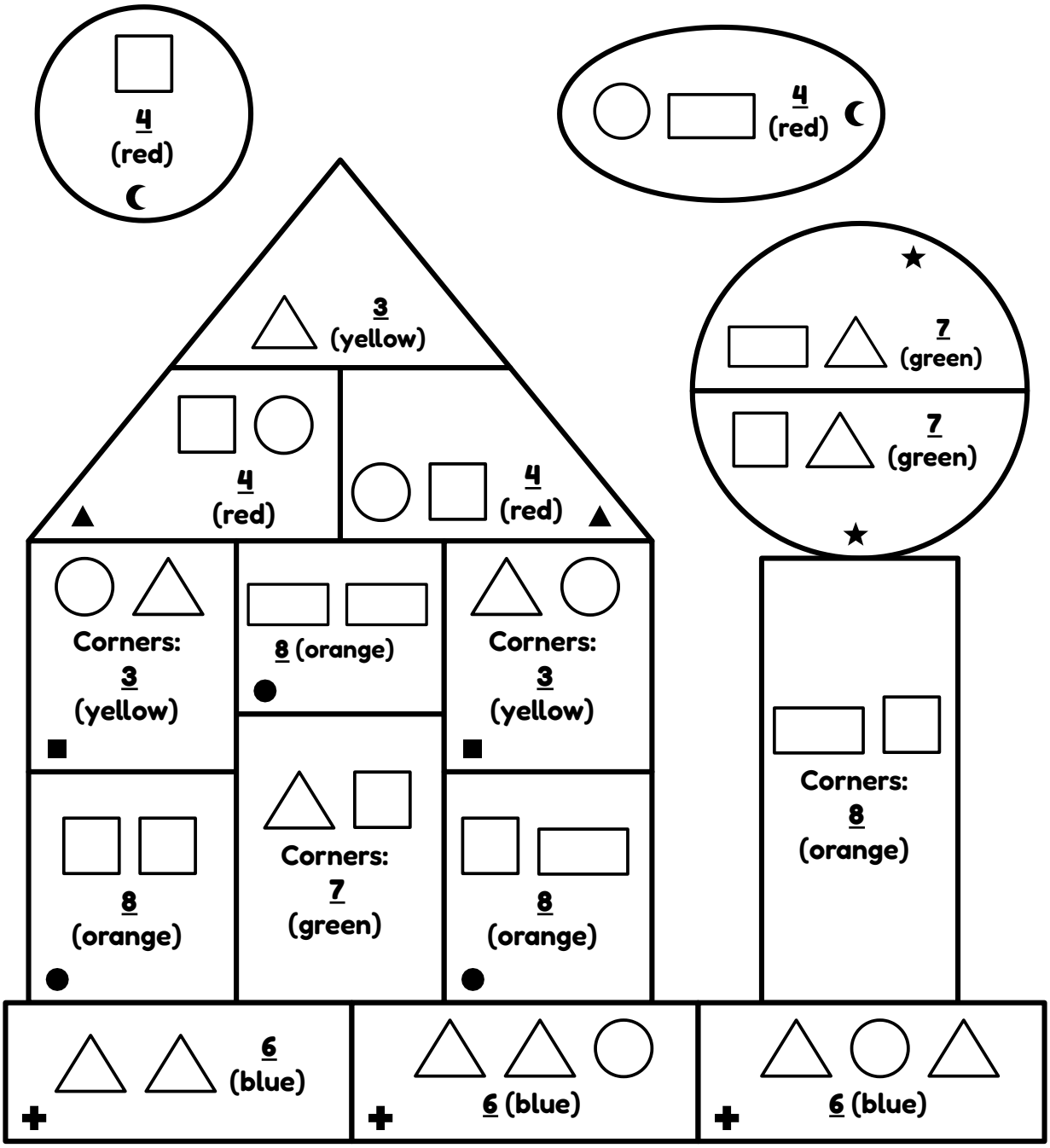
Every part shows the question, the answer underlined and the colour in brackets. Pupils colour their own copy. Colour key: 1 = yellow, 2 = red, 3 = blue, 4 = green, 5 = orange.



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Answers: S1-04 (P1-12) Colour the House

Every part shows the question, the answer underlined and the colour in brackets. Pupils colour their own copy. Colour key: 3 = yellow, 4 = red, 6 = blue, 7 = green, 8 = orange.



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